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Design of an extreme ultraviolet lithography projection objective with a grouping design method through forward and reverse real ray tracing

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As the logic node gets more and more advanced, the performance of extreme ultraviolet (EUV) objective projection is required to be higher and higher in a large field of view. It is known that a good initial structure can greatly reduce the dependence on the experience of optical designers. In this paper, a grouping design method through forward and reverse real ray tracing is proposed to design the aspheric initial structure for the EUV objective system. The system is first divided into three groups, and each spherical group is designed separately. Then, the three groups are connected as a whole spherical initial objective system. Through forward and reverse real ray tracing, each spherical group is recalculated to an aspheric structure in turn. Finally, an iterative process is applied to improve the performance of the aspheric initial structure. The aspheric initial structure calculated by this method can be taken as a good starting point for further optimization. As verification of the design method, a six-aspheric-mirror EUV lithography objective with a numerical aperture of 0.33 has been designed, whose root mean square (RMS) wavefront error is less than 0.2 nm and distortion is less than 0.1 nm. © 2022 Optica Publishing Group

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1. INTRODUCTION

Over the past few years, extreme ultraviolet (EUV) lithography with a numerical aperture (NA) of 0.33 has been employed for 7 and 5 nm nodes [1]. The projection objective is a key component for EUV lithography tools, and it adopts a six-aspheric-mirror reflective structure in industry nowadays [2]. As the technology node gets more and more advanced, the performance of the EUV objective is required to be higher and higher [3]. As the EUV lithography tools evolve from 3300- to 3400-type, the root mean square (RMS) wavefront error of the EUV lithography objective system is reduced from less than 0.4 nm to less than 0.2 nm, and at the same time the distortion is reduced from less than 1 nm to less than 0.5 nm [4,5]. The design of the EUV lithography objective is usually achieved by selecting or calculating an appropriate initial structure and then optimizing it. Thus, a good initial structure can greatly reduce the dependence on the experience of optical designers in the optimization process.

In the last few years, several design methods for the initial structure of an EUV lithography objective have been proposed. Lerner *et al*. applied the Y-y bar method to obtain the initial structure of the four-mirror EUV lithography objective [6]. But this method must know the heights of the chief ray and marginal ray on each mirror in advance, which are difficult to determine. Bal proposed the paraxial search method to obtain the suitable spherical initial structure by performing an exhaustive search of all spherical initial structures, which is quite inefficient [7]. Hudyma *et al*. applied the concept of grouping to design the initial structure of EUV lithography objectives, but the details of this method were not mentioned [8]. In our previous work, we developed the grouping strategy and proposed the grouping design method with real ray tracing [9–12]. Through this method, the spherical initial EUV lithography objectives with no less than six mirrors could be obtained. Hsiao and Lin used the grouping strategy and the generalized Gaussian constants to calculate the initial spherical structure of EUV lithography objectives [13]. Wu *et al*. used the grouping strategy and simulated the annealing particle swarm algorithm to design the spherical initial structural of EUV objectives, which can balance the aberrations for the initial structure [14]. However, existing EUV objectives all use aspheric structures, which means that the spherical initial structure designed by the above methods is far from the final design result. To achieve the higher and higher design requirements of the EUV objective, optical designers must take a lot of time to upgrade the surface type from spherical to aspheric by employing different optimization strategies.

In some cases, one may even fail to obtain a suitable aspherical structure. Hence, it is important to have an aspherical initial structure design method for the EUV lithography objective.

In this paper, a grouping design method through forward and reverse real ray tracing is proposed to design the aspheric initial structure for the EUV objective system. The objective system is first divided into three groups, and each spherical group is designed separately. Then, the three groups as a whole spherical initial structure are connected. Through a forward and reverse real ray tracing in the whole system, each spherical group is recalculated to an aspheric group in turn. Finally, an iterative process is applied to improve the performance of the aspheric initial structure. Compared with the spherical structure, the mean value of the RMS spot size for the aspherical initial structure designed by the above method will be greatly reduced, which can be taken as a good starting point for future optimization. To verify the validity and correctness of this method, a high-performance EUV lithography objective with NA 0.33 was designed.

2. METHOD

A. Grouping Design Process for the Spherical Initial Structure

The grouping design process is generally dividing the complicated optical systems into several lens groups and then designing each group separately. As shown in Fig. 1, the grouping strategy is splitting the system into an object-side group (G1), a middle group (G2), and an image-side group (G3) to reduce the complexity of the design for the initial structure. The projection objective with NA 0.33 adopts a six-aspheric-mirror reflective structure in industry nowadays; thus, each group includes two mirrors.

Many spherical initial structures of the objectives have been designed through this method in our early work [9–12]. The parameters of the object-side group and the image-side group are calculated based on the object-height, the NA, the demagnification, and the distance between the marginal ray and the boundary of each mirror. In the design process, the distance between the marginal ray and the boundary of each mirror is to ensure that the marginal ray is unobscured by the mirrors. The middle group is an important group to connect the object-side group and the image-side group, which must obey the principles of the pupil matching and object-image matching. The parameters of the two conjugate imaging relationships can be obtained through G1 and G3, and based on that, the parameters of G2 can be calculated by Snell's law. Finally, the spherical initial structure of the EUV lithography objective can be obtained through directly connecting the three groups.

B. Grouping Design Method through Forward and Reverse Real Ray Tracing

1. *Design Process for Aspheric Initial Structure for One Lens Group*

Compared with the spherical initial, the aspheric structure will be closer to the final initial structure. In this section, the spherical lens group will be recalculated to an aspheric lens group. Sample field of view (FOV) points A and A′ are the conjugate points on the mask and wafer, respectively. If there is no aberration in the objective system, the rays from A will all reach A′. However, due to the aberration of the initial structure, the intersection of the rays from the ideal FOV point A and the wafer will deviate from the ideal FOV point A′ in the forward path as shown in Fig. 2(a). Similar to that, as shown in Fig. 2(b), the intersection of the rays from the ideal FOV point A′ and the mask will also deviate from the ideal FOV point A in the reverse path.

Thus, due to the aberrations of the objective system, the intersections of the same feature rays, respectively, from the ideal FOV point A and from the ideal FOV point A' with the same mirror will be different. As shown in Fig. 3, one lens group is composed of two mirrors in the forward path, which are named as the first mirror and the second mirror, respectively. The feature ray $\vec{R}$ is from the ideal FOV point A on the object space, and the intersection P is the intersection of this feature ray and the first mirror in the forward path. The ray $\vec{R'}$ is the same feature ray as $\vec{R}$, which is from the ideal FOV point A' on the image space, and the intersection P' is the intersection of this feature ray and the second mirror in the reverse path. If there are no aberrations of this system, the ray $\vec{R}$ is redirected into $\overrightarrow{PP'}$ by the first mirror, and then the ray $\overrightarrow{PP'}$ will be redirected into $\vec{R'}$ in the forward path. The unit normal vector $\vec{N}$ at the point P on the first mirror can be calculated based on Snell's law.

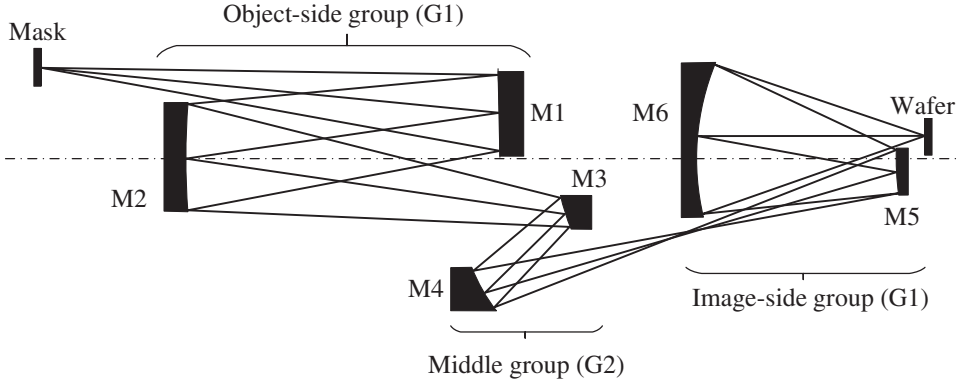


Fig. 1. Grouping strategy for the EUV lithography objective system.

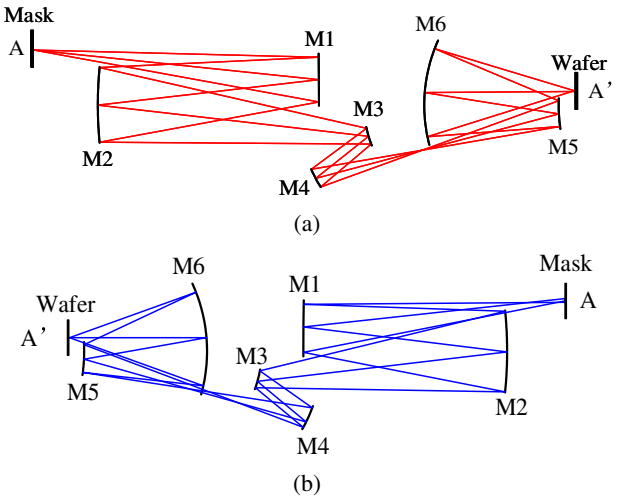


Fig. 2. Ray path of the initial spherical system (a) in the forward path and (b) in the reverse path.

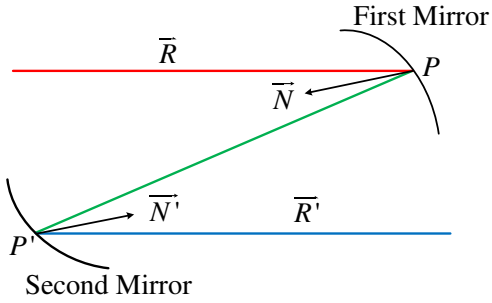


Fig. 3. Schematic overview of the method to calculate one aspheric lens group.

Furthermore, the unit normal vector $\vec{N'}$ at the point P' on the second mirror can also be calculated based on Snell's law.

2. Aspheric Surface Fitting

As shown in Fig. 4, through the above design process, the intersections of each mirror and different feature rays over the entire aperture and full FOV can be calculated on the same global coordinate system. Then through Snell's law, the normal vector $\vec{N} = (U, V, -1)$ at each intersection point P on the first mirror can be calculated, and the normal vector $\vec{N'} = (U', V', -1)$ at each intersection point P' on the second mirror can also be calculated on the same global coordinate system.

These intersection points are fitted to a rotationally symmetric aspheric surface, in which the origin is at $(0, 0, Z_0)$ and the rotational symmetric axis is the Z axis on the global coordinate system. In the local coordinate system, the expression of an even aspheric surface defined under the local coordinate system can be written as

$$Z = \frac{cr^2}{1 + \sqrt{1 - (1+k)(c^2 r^2)}} + \sum_{i=2}^{n} a_i r^{2i}, \tag{1}$$

where c is the curvature at the pole of the surface, k is the conic constant, a_i coefficients, r is the radial distance, and $r^2 = x^2 + y^2$.

Thus, in the global coordinate system, the expression of an even aspheric surface is

$$Z = \frac{cr^2}{1 + \sqrt{1 - (1+k)(c^2 r^2)}} + \sum_{i=2}^{n} a_i r^{2i} + Z_0. \tag{2}$$

Through the Taylor series expansion, Eq. (2) can be transformed into linear polynomials,

$$Z = \sum_{i=0}^{n} b_i r^{2i}. \tag{3}$$

The expression for the coefficient b_i is

$$b_i = \begin{cases} Z_o & (i=0) \\ \frac{c}{2} & (i=1) \\ \left(\frac{(2i)!(1+k)^{i-1}c^{2i-1}}{2^{2i}(i!)^2(2i-1)}\right) + a_n & (i \geq 2) \end{cases}. \tag{4}$$

Then, according to Eqs. (1)–(4), the parameters and origin coordinate of first aspheric mirror surface can be calculated by the method in [15], considering both coordinates P and normal $\vec{N} = (U, V, -1)$ of discrete data points. Similar to the fitting process for the first mirror, the second mirror can also be fitted to the aspheric mirror.

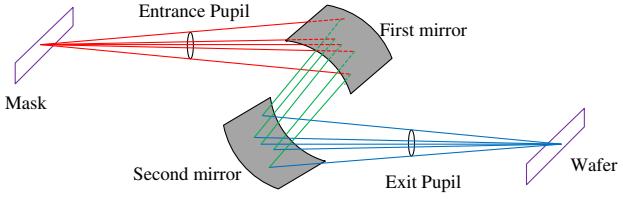


Fig. 4. Calculation process of one aspheric lens group with different feature rays from full FOV points.

3. Iterative Process

With the method described in Section 2.B, one group can be fitted into the aspheric structure based on the real ray tracing in the forward path and the reverse path. In this section, an iterative process will be applied to further improve the imaging quality of the initial structure. During the iterative process, the mean value of the RMS spot size (MRSS) of all FOV points for the forward system will be adopted as the merit value to evaluate the performance after the parameters of each group are recalculated. The value of the RMS spot size for each sample FOV can be obtained by the CODE V software. Through the forward and reverse real ray tracing in the whole system, each spherical group is recalculated to an aspheric structure in turn, and each time after calculating, the previous group will be replaced by the new group, and the new system is taken as the new initial structure for the next design process.

As shown in Fig. 5, the merit value of the forward system will be evaluated each time when the parameters of each group are recalculated. The ending requirement for the iteration process in our design is defined as the absolute value of difference between the current merit value and that of the previous round being less than the threshold ε. When the current aspheric

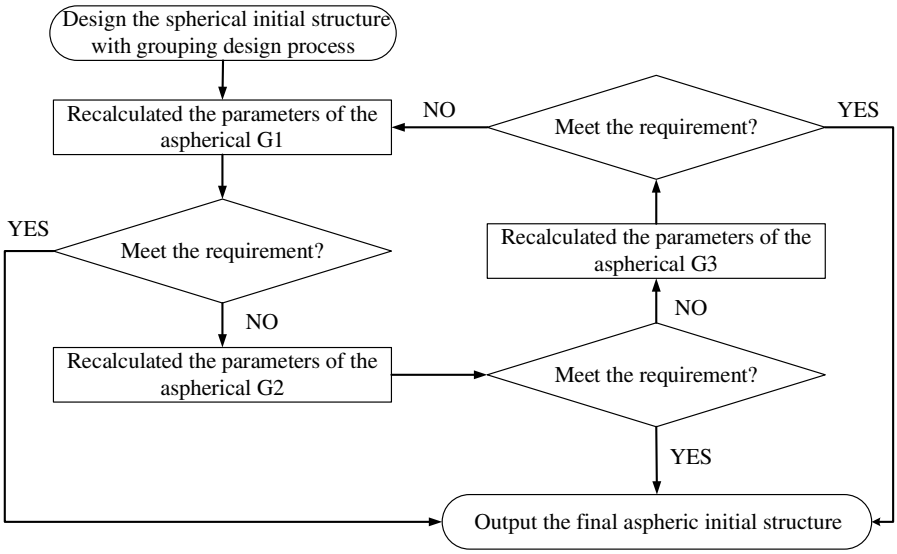


Fig. 5. Flow diagram of the grouping design method through forward and reverse real ray tracing for the EUV lithography objective system.

system meets the ending requirement, the current system will be taken as the final initial aspheric structure for further optimization.

3. OPTIMIZATION AND PERFORMANCE

As shown in Fig. 6, a spherical initial structure of the EUV lithography objective with NA 0.33 can be obtained through the grouping design process described in Section 2.A. The mean value of the RMS spot size of all FOV points for the objective is 1.157 mm.

To further reduce the aberrations of the initial structure, the spherical initial structure shown in Fig. 6 will be recalculated to an aspheric initial structure by the method described in Section 2.B. In the grouping design method through forward and reverse real ray tracing, 20 sample fields over the half-full FOV are employed in the surface recalculation process. The aperture of each field is divided into seven angles with equal interval. Seven different pupil coordinates are sampled along the radial direction of each angle. Thus, a total of 1000 feature rays are used in the recalculation process, and after 100 rounds of the iterative process, all the mirrors of the initial structure are finally fitted into even aspheric surfaces. The change of the mean value of the RMS spot size for the system during the iterative process is shown in Fig. 7. In the design, the ε can be set as 0.0001 mm. The merit value that meets the requirement is at the iteration number 43. However, in this paper, to better show the following

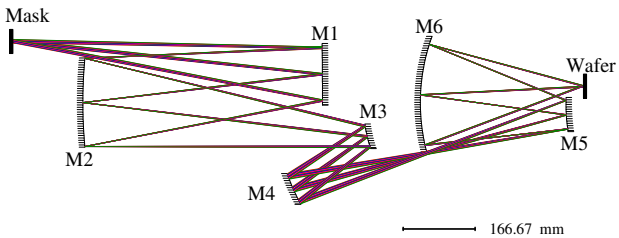


Fig. 6. Spherical initial structure of the EUV lithography objective with NA 0.33.

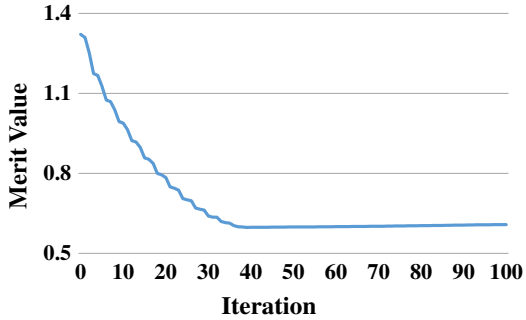


Fig. 7. Change of the mean value of the RMS spot size for the system during the iterative process.

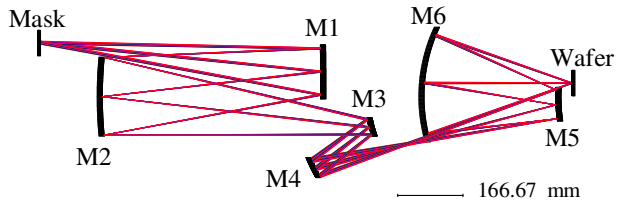


Fig. 8. Aspheric initial structure of the EUV lithography objective with NA 0.33.

change trend of the merit value, 100 rounds of merit value were shown in Fig. 7.

Figure 8 shows the aspheric initial structure designed by the grouping design method through the forward and reverse real ray tracing design process. The fitted coefficients of each aspheric mirror are described in Table 1. Figure 9 shows the mean value of the RMS spot size for the initial structure of the spherical initial structure and the aspheric initial structure, respectively. Compared to the spherical initial structure, the mean value of the RMS spot size for the aspheric initial structure designed by the grouping design method through the forward and reverse real ray tracing design process will decrease by

Table 1. Fitted Coefficients for Each Aspheric Mirror of the Aspheric Initial Structure

Surface	Curvature	k	a_2	a_3	a_4	a_5	a_6
M1	−1.68E-04	0	−7.74E-11	5.65E-15	6.80E-18	−7.63E-22	2.23E-26
M2	9.17E-04	0	−7.74E-11	5.65E-15	6.80E-18	−7.63E-22	2.23E-26
M3	3.49E-03	0	−7.74E-11	5.65E-15	6.80E-18	−7.63E-22	2.23E-26
M4	2.42E-03	0	8.62E-09	−9.66E-13	4.01E-17	−7.32E-22	4.95E-27
M5	2.36E-03	0	1.84E-08	−3.48E-11	1.67E-14	−2.22E-18	0
M6	2.42E-03	0	1.59E-10	−5.91E-14	7.26E-18	−3.66E-22	6.47E-27

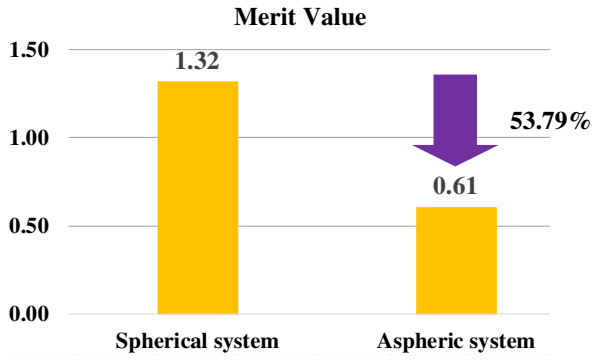


Fig. 9. Mean value of the RMS spot size for the spherical initial structure and the aspheric initial structure, respectively.

53.79%, which means that the aspheric initial structure is closer to the final structure.

A further optimization process will be applied to achieve a high imaging quality with low aberrations. The layout of the EUV lithography objective with NA 0.33 after further optimization is shown in Fig. 10.

The optical characteristics of the EUV lithography objective with NA 0.33 are described in Table 2. The total track is 1433.10 mm. The telecentricity of the EUV lithography objective on the image side is less than 0.17°. The chief ray angle

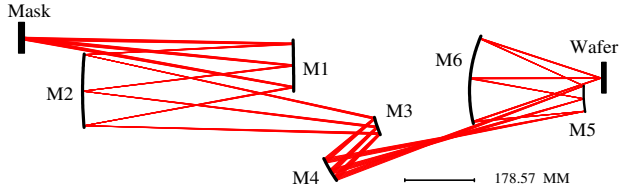
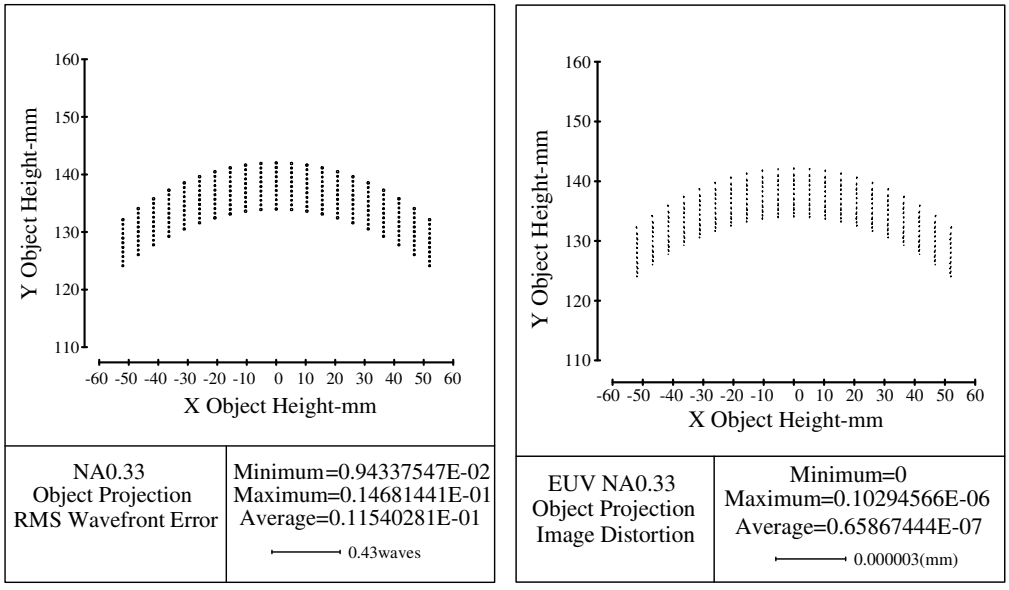


Fig. 10. Layout of the final EUV lithography objective with NA 0.33.

Table 2. Optical Characteristics of the Final EUV Lithography Objective with NA 0.33

Parameter	Specification
Wavelength	13.5 nm
Numerical aperture	0.33
Demagnification	4
Image-side field of view	26 mm × 2 mm (arc)
Total track length	1433.10 mm
Chief ray angle on mask	≤6°
Telecentricity	≤0.17°
Wavefront error (RMS)	≤0.1982 nm
Max distortion	≤0.102 nm

on the mask is less than 6°. The RMS wavefront error of the EUV objective is less than 0.1982 nm across a 26 mm × 2 mm arc-shaped field as shown in Fig. 11(a), and the maximum



(a) (b)

Fig. 11. (a) Composite RMS wavefront error and (b) distortion of the EUV lithography objective with NA 0.33.

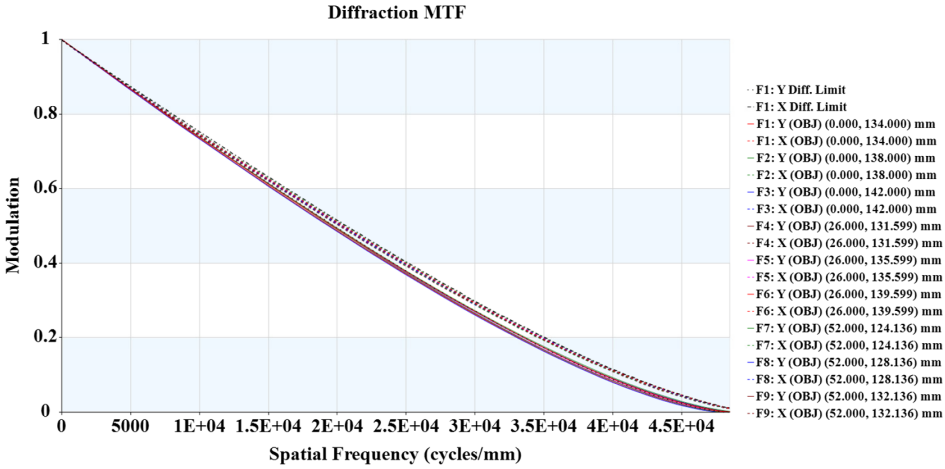


Fig. 12. Modulation transfer function of the EUV lithography objective with NA 0.33.

distortion is less than 0.102 nm as shown in Fig. 11(b). The RMS wavefront error and the distortion of the EUV lithography objective system meet the low aberration design requirements. Furthermore, the modulation transfer function (MTF) of the EUV lithography objective is shown in Fig. 12, which indicates that the image quality is close to the diffraction limitation.

4. CONCLUSION

The grouping design method through forward and reverse real ray tracing is proposed to design the EUV lithography objective system. First, the objective system is divided into three groups, and the spherical parameters of each group can be obtained through Snell's law. Then, the three groups will be connected as the spherical initial structure. Through forward and reverse real ray tracing and Snell's law, each group will be recalculated to an aspheric group in turn. Finally, an iterative process is adopted to improve the performance of the aspheric initial structure. A six-aspheric-mirror EUV lithography objective with NA 0.33 was designed by the above method. Compared to the spherical initial structure, the mean value of the RMS spot size for the aspheric initial structure after the iteration process decreases by 53.79%. Thus, the aspheric initial structure that is closer to the final structure can be taken as a better starting point for future optimization. Through the optimization process, a high-performance objective system with low aberrations can be obtained. The MTF of the EUV lithography objective is close to the diffraction limit. The telecentricity of the EUV lithography objective on the image side is less than 0.17°. The chief ray angle on the mask is less than 6°. The RMS wavefront error of the EUV lithography objective is less than 0.1982 nm across a 26 mm × 2 mm arc-shaped field, and the distortion is less than 0.102 nm, which means that the EUV lithography objective system meets the low aberration design requirements.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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